

1. Package `studentinfo` – Student Grade

Student.java

```
package studentinfo;

import java.util.Scanner;

public class Student {
    int studentId;
    String studentName;
    String course;
    double marks;

    public void acceptDetails() {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Student ID: ");
        studentId = sc.nextInt();
        sc.nextLine();

        System.out.print("Enter Student Name: ");
        studentName = sc.nextLine();

        System.out.print("Enter Course: ");
        course = sc.nextLine();

        System.out.print("Enter Marks: ");
        marks = sc.nextDouble();
    }

    public String calculateGrade() {
        if (marks >= 90)
            return "A+";
        else if (marks >= 80)
            return "A";
        else if (marks >= 70)
            return "B";
        else if (marks >= 60)
            return "C";
        else
            return "D";
    }

    public void displayDetails() {
        System.out.println("Student ID: " + studentId);
        System.out.println("Student Name: " + studentName);
        System.out.println("Course: " + course);
        System.out.println("Marks: " + marks);
        System.out.println("Grade: " + calculateGrade());
    }
}
```

StudentTest.java

```

import studentinfo.Student;

public class StudentTest {
    public static void main(String[] args) {
        Student s = new Student();

        s.acceptDetails();
        System.out.println();

        s.displayDetails();
    }
}

```

2. Package `company` and `department` – Manager Salary

Employee.java

```

package company;

public interface Employee {
    void calculateSalary();
}

```

Manager.java

```

package department;

import company.Employee;
import java.util.Scanner;

public class Manager implements Employee {
    String name;
    double basicSalary;
    double salary;

    public void acceptDetails() {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Manager Name: ");
        name = sc.nextLine();

        System.out.print("Enter Basic Salary: ");
        basicSalary = sc.nextDouble();
    }

    public void calculateSalary() {
        salary = basicSalary + (basicSalary * 20 / 100);
    }

    public void display() {
        System.out.println("Manager Name: " + name);
        System.out.println("Salary: " + salary);
    }
}

```

ManagerTest.java

```
import department.Manager;

public class ManagerTest {
    public static void main(String[] args) {
        Manager m = new Manager();

        m.acceptDetails();
        m.calculateSalary();
        m.display();
    }
}
```

3. Employee Interface – Manager and Developer

```
import java.util.Scanner;

interface Employee {
    void calculateSalary();
    void displayDetails();
}

class Manager implements Employee {
    String name;
    double salary;

    Manager(String name, double salary) {
        this.name = name;
        this.salary = salary;
    }

    public void calculateSalary() {
        salary = salary + (salary * 20 / 100);
    }

    public void displayDetails() {
        System.out.println("Manager Name: " + name);
        System.out.println("Salary: " + salary);
    }
}

class Developer implements Employee {
    String name;
    double salary;

    Developer(String name, double salary) {
        this.name = name;
        this.salary = salary;
    }

    public void calculateSalary() {
        salary = salary + (salary * 10 / 100);
    }
}
```

```

        public void displayDetails() {
            System.out.println("Developer Name: " + name);
            System.out.println("Salary: " + salary);
        }
    }

    public class EmployeeTest {
        public static void main(String[] args) {
            Scanner sc = new Scanner(System.in);

            System.out.print("Enter Manager Name: ");
            String managerName = sc.nextLine();

            System.out.print("Enter Manager Salary: ");
            double managerSalary = sc.nextDouble();

            sc.nextLine();

            System.out.print("Enter Developer Name: ");
            String developerName = sc.nextLine();

            System.out.print("Enter Developer Salary: ");
            double developerSalary = sc.nextDouble();

            Manager m = new Manager(managerName, managerSalary);
            Developer d = new Developer(developerName, developerSalary);

            m.calculateSalary();
            d.calculateSalary();

            System.out.println();

            m.displayDetails();

            System.out.println();

            d.displayDetails();
        }
    }
}

```

4. Interface for Bank Account

```

import java.util.Scanner;

interface Account {
    void deposit(double amount);
    void withdraw(double amount);
    void checkBalance();
}

class SavingsAccount implements Account {
    double balance;

    SavingsAccount(double balance) {
        this.balance = balance;
    }
}

```

```

    }

    public void deposit(double amount) {
        balance = balance + amount;
        System.out.println("Amount deposited");
    }

    public void withdraw(double amount) {
        if (amount <= balance) {
            balance = balance - amount;
            System.out.println("Amount withdrawn");
        } else {
            System.out.println("Insufficient balance");
        }
    }

    public void checkBalance() {
        System.out.println("Balance: " + balance);
    }
}

public class BankInterface {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter initial balance: ");
        double balance = sc.nextDouble();

        SavingsAccount a = new SavingsAccount(balance);

        System.out.print("Enter amount to deposit: ");
        double deposit = sc.nextDouble();
        a.deposit(deposit);

        System.out.print("Enter amount to withdraw: ");
        double withdraw = sc.nextDouble();
        a.withdraw(withdraw);

        a.checkBalance();
    }
}

```